

LinguaFlow - Complete Updated Tech Stack (Verified)

Verified against:

- `pubspec.yaml`
- `pubspec.lock` (resolved versions)
- Key implementation files in `lib/services` and `lib/screens`

Frontend

- Flutter: $\geq 3.38.4$ (SDK constraint from lockfile)
- Dart: $\geq 3.11.5 < 4.0.0$
- flutter_riverpod: 2.6.1
- go_router: 14.8.1
- fl_chart: 0.68.0

Backend and Database

- firebase_auth: 5.7.0
- cloud_firestore: 5.6.12
- firebase_messaging: 15.2.10
- sqflite: 2.4.2+1

AI and Machine Learning

- Groq-compatible chat completions endpoint via `OpenAiService`
- Endpoint: `https://api.groq.com/openai/v1/chat/completions`
- Model: `llama-3.1-8b-instant`
- SM-2 style spaced repetition algorithm (`calculateSrs`)
- Levenshtein distance for pronunciation similarity scoring
- MediaPipe hand models:
- `assets/models/hand\_landmark.tflite`
- `assets/models/palm\_detection.tflite`
- Geometric/rule-based ASL letter classification in sign detector logic

Speech and Audio

- speech_to_text: 7.3.0
- flutter_tts: 4.2.5

Computer Vision

- camera: 0.10.6
- tflite_flutter: 0.11.0
- image: 4.8.0

Document Generation

- pdf: 3.12.0
- printing: 5.14.3
- intl: 0.19.0

Storage and Preferences

- sqflite: 2.4.2+1
- shared_preferences: 2.5.5
- path_provider: 2.1.5

Notifications

- flutter_local_notifications: 17.2.4
- firebase_messaging: 15.2.10
- timezone: 0.9.4

Networking and APIs

- http: 1.6.0
- cached_network_image: 3.4.1

Architecture and Patterns in Codebase

- Service-oriented architecture (service classes for auth, lessons, quiz, vocab, notifications, certificate, chatbot)
- Offline-first lessons flow (local SQLite + Firestore sync path)
- Seeder pattern (`SeederService`) for lessons/quizzes/sign lessons
- Singleton usage where applicable (e.g., NotificationService, LocalStorageService)
- SRS-based vocabulary review scheduling

Core Firestore Collections (Observed)

- `users/{uid}`
- `users/{uid}/vocabulary`
- `users/{uid}/quizResults`
- `users/{uid}/chatHistory`
- `lessons`
- `quizzes`
- `sign\_lessons`

Accuracy Corrections Compared to Provided Draft

- Corrected package versions to resolved lockfile values where they differed:
 - `speech\_to\_text`: 7.3.0 (not 7.0.0)
 - `camera`: 0.10.6 (not 0.10.5+9)
 - `image`: 4.8.0 (not 4.1.7)
 - `cached\_network\_image`: 3.4.1 (not 3.3.1)
 - `path\_provider`: 2.1.5 (not 2.1.2)
 - `sqflite`: 2.4.2+1 (not 2.4.2 exact)
- Preserved accurate entries that already matched lockfile:
 - `firebase\_auth` 5.7.0
 - `cloud\_firestore` 5.6.12
 - `firebase\_messaging` 15.2.10
 - `flutter\_local\_notifications` 17.2.4
 - `go\_router` 14.8.1
 - `flutter\_riverpod` 2.6.1
 - `pdf` 3.12.0
 - `printing` 5.14.3
 - `shared\_preferences` 2.5.5
 - `http` 1.6.0
 - `timezone` 0.9.4